

Immersive Gallery

Perceptual and Feature-Driven Navigation

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Project Overview

The goal of this project is to develop an interactive 3D gallery where images and videos are organised according to perceptual visual features. Instead of a static layout, the system maps each media element into a spatial structure derived from its visual properties, enabling exploration through movement and interaction.

The system can initially be prototyped as a 2D gallery to validate feature extraction and interaction concepts, before extending to a fully immersive 3D environment.

The gallery is presented as a dark immersive environment populated by floating frames. Media elements are arranged using a cylindrical colour-based mapping, creating a perceptually coherent structure where similar items are positioned close to each other.

To address visual occlusion and improve navigation, the system introduces a radial exploration mechanism. Rather than displaying all elements with equal emphasis, only a subset located around a selected radial layer is highlighted and projected toward the visible boundary. This allows users to explore the dataset through successive layers without losing spatial context.

The application will be implemented using openFrameworks (C++) and will support both traditional input (mouse and keyboard) and camera-based interaction.

Interface Sketch and Storyboard

The interface is organised into four main states, illustrated in Annex A.

Screen 1 — Default 3D Gallery View

The system starts in a 3D environment where media frames are distributed around the user. The space is dark, with subtle lighting highlighting nearby elements. The camera is positioned at eye level and can be freely controlled.

Only elements belonging to the current radial layer are fully visible, while others appear faded to preserve context and reduce clutter.

Screen 2 — Focus and Metadata Overlay

When the user approaches or hovers over a frame, it is highlighted and slightly enlarged. A floating panel appears displaying metadata such as colour properties and feature values.

Screen 3 — Fullscreen View



Figure 1: Annex A

Selecting a media item opens it in fullscreen. Images fill the display, while videos play automatically and can be paused or resumed. A minimal overlay provides controls and basic metadata.

Screen 4 — Radial Navigation and Reorganisation

Users can explore the gallery by changing the active radial layer, revealing different subsets of media. Radial navigation can be controlled either with mouse scroll or with horizontal swipe gestures, where swiping left or right performs the same function as scrolling through the radial layers.

User Interface Elements

The interface consists of a navigable 3D environment where media elements are organised according to an underlying cylindrical mapping.

Navigation is performed using free camera controls, allowing rotation, translation, and zoom. Vertical movement enables the exploration of different luminance levels.

Radial navigation is controlled through scroll input. In addition, touch interaction is supported through horizontal swipe gestures (left/right), which replicate the behaviour of mouse scrolling by changing the active radial layer.

Media frames are rendered as textured quads. Lighting effects, such as spotlights, enhance depth perception and focus.

To mitigate occlusion, radial layer control is used. The user adjusts the active radial level using scroll or swipe input. Media elements close to this level are emphasised and projected toward the visible boundary, while others remain in the background with reduced visibility.

When interacting with a media item, a floating metadata panel appears. Selecting an item opens it in fullscreen mode.

Perceptual Features and Behaviour

Visual features are extracted from each media item at load time and stored in XML format. These values are used consistently for spatial mapping, similarity computation, and behaviour.

The spatial organisation is based on the HSV colour model:

- Hue defines angular position
- Saturation defines radial distance
- Luminance defines vertical placement

This mapping defines the global spatial structure of the gallery.

Additional features control visual and dynamic behaviours without altering the overall spatial organisation. Luminance variance affects visual emphasis, such as glow or border intensity. Edge density influences perceived visual detail and responsiveness. The number of keypoints controls the size of each media element.

For videos, motion energy is computed using frame differences and drives animation intensity or dynamic visual effects.

At least three distinct features are used to control different behaviours: luminance variance for emphasis, edge density for visual complexity, and keypoint count for size scaling.

Similarity and Spatial Behaviour

Similarity between media items is primarily determined by the spatial mapping defined by the extracted features. Each element is positioned in a cylindrical coordinate system derived from the HSV colour model, where hue defines angular position, saturation defines radial distance, and luminance defines vertical placement.

This mapping ensures that visually similar items naturally occupy nearby regions in space. Media with similar colour distributions are placed at similar angular positions, while comparable saturation and luminance values result in similar radial and vertical placement. As a result, visually related items emerge in neighbouring areas of the gallery through the spatial organisation itself.

The radial navigation mechanism further reinforces this behaviour. Since saturation controls radial distance, users explore the gallery through layers of colour intensity. Within each radial layer, items share similar saturation values and are therefore more likely to exhibit perceptual similarity, making navigation between related elements intuitive and continuous.

Overall, similarity is not imposed as a separate clustering process, but emerges from the combination of perceptual mapping and feature-driven refinement, enabling both automatic grouping and natural exploration of related media.

Preliminary Class Structure

The system is organised into several components.

The **MediaItem** class stores data related to each media element, including features and spatial position.

The **FeatureExtractor** computes visual features such as colour, texture, and motion.

The **SpatialMapper** converts feature values into spatial coordinates.

The **SimilarityEngine** computes distances between media items and supports similarity-based refinement.

The **InteractionManager** handles user input, including gestures and camera-based interaction.

The **Renderer** manages drawing and visual effects.

The **MetadataManager** handles XML storage and retrieval.

Presence and Immersion

Several techniques are used to enhance immersion and spatial perception within the abstract gallery environment.

The virtual camera is free-floating and supports smooth motion with inertia, allowing continuous exploration of the cylindrical space. This movement model reinforces the perception of navigating through a structured visual field rather than a fixed viewpoint.

Lighting effects dynamically highlight nearby media elements, while depth cues such as fog and distance-based attenuation reinforce spatial relationships and reduce visual clutter.

Gesture-based interaction and camera input further enhance the experience, enabling intuitive navigation through swipe-based radial control and motion-driven interaction.

Conclusion

This project proposes an interactive gallery where spatial organisation is driven by perceptual features. By combining a colour-based mapping with radial exploration and feature-driven behaviour, the system enables intuitive navigation and meaningful grouping within an immersive 3D environment.